

Detailed Breakdown of Question 3c

2021 ORF309 Midterm 1

1 Problem Statement

Find the probability that you get exposed to COVID **before** receiving your first dose of the vaccine.

2 Key Random Variables

Random Variables

1. X_i : Indicator random variable for COVID exposure on day i
 - $P(X_i = 1) = 0.01$ (exposed on day i)
 - $P(X_i = 0) = 0.99$ (not exposed on day i)
2. N : The day you receive your first vaccine dose
 - $N \sim \text{Geometric}(0.05)$
 - Each day you have a 5% chance of getting a vaccine slot
3. A : The event that you do **NOT** get exposed to COVID before the first dose

3 Solution Strategy

The solution uses the **Law of Total Probability**, conditioning on when you receive the vaccine ($N = n$).

3.1 Why Condition on N ?

Since N is a *random variable* (you don't know in advance when you'll get the vaccine), we must consider all possible days $n = 1, 2, 3, \dots$ and weight each by its probability.

4 Step-by-Step Calculation

4.1 Step 1: Express probability for a specific day n

Using the Law of Total Probability:

$$P(A) = \sum_{n=1}^{\infty} P\left(\sum_{i=1}^{n-1} X_i = 0 \mid N = n\right) \cdot P(N = n) \quad (1)$$

The term $\sum_{i=1}^{n-1} X_i = 0$ means you were **NOT** exposed on any day from 1 to $n - 1$.

4.2 Step 2: Use independence

Since X_i and N are independent events:

$$P\left(\sum_{i=1}^{n-1} X_i = 0 \mid N = n\right) = P\left(\sum_{i=1}^{n-1} X_i = 0\right) = 0.99^{n-1} \quad (2)$$

This is because you need to avoid exposure for $(n - 1)$ consecutive days, each with probability 0.99.

4.3 Step 3: Apply Geometric distribution for N

For a Geometric(p) random variable with parameter $p = 0.05$:

$$P(N = n) = 0.05 \cdot 0.95^{n-1} \quad (3)$$

4.4 Step 4: Combine and simplify

Substituting into the equation from Step 1:

$$P(A) = \sum_{n=1}^{\infty} 0.99^{n-1} \cdot 0.05 \cdot 0.95^{n-1} \quad (4)$$

$$= \sum_{n=1}^{\infty} 0.05 \cdot (0.99 \cdot 0.95)^{n-1} \quad (5)$$

$$= 0.05 \cdot \sum_{n=1}^{\infty} (0.9405)^{n-1} \quad (6)$$

4.5 Step 5: Use geometric series formula

For $|r| < 1$, the geometric series formula gives:

$$\sum_{n=1}^{\infty} r^{n-1} = \frac{1}{1 - r} \quad (7)$$

Applying this with $r = 0.9405$:

$$P(A) = \frac{0.05}{1 - 0.95 \cdot 0.99} \quad (8)$$

$$= \frac{0.05}{1 - 0.9405} \quad (9)$$

$$= \frac{0.05}{0.0595} \quad (10)$$

$$\approx 0.8403 \quad (11)$$

4.6 Final Answer

The probability of getting exposed **before** the first dose is:

Final Answer

$$P(\text{exposed before first dose}) = 1 - P(A) = 1 - \frac{0.05}{1 - 0.95 \cdot 0.99} \approx 0.1597 \quad (12)$$

Or equivalently: $1 - \frac{0.05}{1 - 0.95 \cdot 0.99}$

5 Key Insights

1. **Why condition on N ?** Because N is random—you don't know when you'll get the vaccine, so you must account for all possible days.
2. **Why use independence?** X_i (daily COVID exposure) and N (vaccine arrival) are independent events, which allows us to separate the conditional probability.
3. **Physical interpretation:** There's roughly an 84% chance you avoid COVID exposure before getting your first vaccine dose, which means about a 16% chance you get exposed before being vaccinated.

6 Common Mistakes (from Grading Comments)

- **Treating N as fixed** (loses 4.0 points): Some students fixed the date $N = n$ and computed the probability only for that specific n , without recognizing that N itself is a random variable.
- **Not using Law of Total Probability:** This is essential when dealing with a random variable that affects the outcome.
- **Computational errors:** Mistakes in applying the geometric series formula or combining terms.